

The third row is $[0 \ 0 \ 0 \ b_1 + b_3 - b_2/2]$. By Theorem 2, the system of equations is inconsistent if and only if $b_1 + b_3 - b_2/2 \neq 0$.

∴ The equation $Ax=b$ is inconsistent for all such b where $b_1 + b_3 - b_2/2 \neq 0$.

□

• In the next theorem, the sentence "The columns of A span $\mathbb{R}^m$ " means that every b in $\mathbb{R}^m$ is a linear combination of the columns of A . In general, a set of vectors $\{v_1, \dots, v_p\}$ in $\mathbb{R}^m$ spans (or generates) $\mathbb{R}^m$ if every vector in $\mathbb{R}^m$ is a linear combination of $v_1, \dots, v_p$ — that is, if $\text{Span}\{v_1, \dots, v_p\} = \mathbb{R}^m$.

Theorem 4: Let A be an $m \times n$ matrix. Then the following statements are logically equivalent. That is, for a particular A , either they are all true or they are all false.

- a) For each b in $\mathbb{R}^m$, the equation $Ax=b$ has a solution.
- b) Each b in $\mathbb{R}^m$ is a linear combination of the columns of A .
- c) The columns of A span $\mathbb{R}^m$.
- d) A has a pivot position in every row.

⊆ v. imp

Statements (a), (b) and (c) are equivalent because of the definition of $Ax=b$ and what it means for a set of vectors to span $\mathbb{R}^m$.

The Equation $Ax=b$ in Example 3 fails to be consistent for all b because the echelon form of A has a row of zeros. If A had a pivot in all three rows, we would not care about the calculations in the augmented column because in this case an echelon form of the augmented matrix could not have a row such as $[0 \ 0 \ 0 \ 1]$.

⊗ This suggests why (a) and (d) are equivalent. A formal proof is given later.